

$\mathcal{K}_0^{\#1}$		$\mathcal{K}_0^{\#1}{}_{\alpha\beta\chi}$					
$\mathcal{K}_0^{\#1} \dagger$	$\frac{1}{2} (-3 \Upsilon_1 k^2 - 3 \overset{(2)}{\mathcal{K}}_3)$	0					
$\mathcal{K}_0^{\#1} \dagger^{\alpha\beta\chi}$	0	$3 k^2 \overset{(4)}{\mathcal{K}}_{15} + 3 \overset{(2)}{\mathcal{K}}_1$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$	
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\Upsilon_2 k^2 + 2 \overset{(2)}{\mathcal{K}}_1$	$\frac{\Upsilon_2 k^2 + 2 \overset{(2)}{\mathcal{K}}_1}{\sqrt{2}}$	0	0		
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\Upsilon_2 k^2 + 2 \overset{(2)}{\mathcal{K}}_1}{\sqrt{2}}$	$\frac{1}{2} (\Upsilon_2 k^2 + 2 \overset{(2)}{\mathcal{K}}_1)$	0	0		
	$\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	$-k^2 \overset{(4)}{\mathcal{K}}_2 - \overset{(2)}{\mathcal{K}}_3$	$\frac{k^2 \overset{(4)}{\mathcal{K}}_2 + \overset{(2)}{\mathcal{K}}_3}{\sqrt{2}}$		
	$\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	$\frac{k^2 \overset{(4)}{\mathcal{K}}_2 + \overset{(2)}{\mathcal{K}}_3}{\sqrt{2}}$	$\frac{1}{2} (-k^2 \overset{(4)}{\mathcal{K}}_2 - \overset{(2)}{\mathcal{K}}_3)$		
					$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$	
					$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0
					$\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0

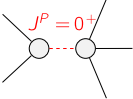
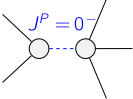
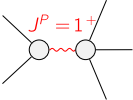
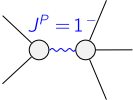
		$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{J}_{0^+}^{\#1} \dagger$		$\frac{1}{\text{Det}(0^+)}$	0				
$\mathcal{J}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$		0	$\frac{1}{3(k^2 \overset{(4)}{\kappa}_{15} + \overset{(2)}{\kappa}_1)}$	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$
	$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$\frac{2}{3 \text{Det}(1^+)}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	0	0		
	$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{1}{3 \text{Det}(1^+)}$	0	0		
	$\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha}$	0	0	$\frac{2}{3 \text{Det}(1^+)}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$		
	$\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha}$	0	0	$-\frac{\sqrt{2}}{3 \text{Det}(1^+)}$	$\frac{1}{3 \text{Det}(1^+)}$	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
				$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0	
				$\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	

Abbreviations used in matrices

$$\Upsilon_1 == \overset{(4)}{\mathcal{K}}_1 + \overset{(4)}{\mathcal{K}}_2 \ \&\& \ \Upsilon_2 == 2 \overset{(4)}{\mathcal{K}}_{15} - \overset{(4)}{\mathcal{K}}_3 \ \&\& \ \&\& \\ \text{Det}(0^+) == -\frac{3}{2} (k^2 (\overset{(4)}{\mathcal{K}}_1 + \overset{(4)}{\mathcal{K}}_2) + \overset{(2)}{\mathcal{K}}_3) \ \&\& \ \text{Det}(1^+) == k^2 (3 \overset{(4)}{\mathcal{K}}_{15} - \frac{3 \overset{(4)}{\mathcal{K}}_3}{2}) + 3 \overset{(2)}{\mathcal{K}}_1$$

Lagrangian

$$\overset{(2)}{\mathcal{K}}_1 \mathcal{K}_{\alpha\beta\chi} \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(2)}{\mathcal{K}}_1 \mathcal{K}^{\alpha\beta\chi} \mathcal{K}_{\beta\alpha\chi} + \overset{(2)}{\mathcal{K}}_3 \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta \mathcal{K}^\chi{}_\chi{}^\beta{}_\beta + \\ \overset{(4)}{\mathcal{K}}_1 \partial_\beta \mathcal{K}^\delta{}_\chi{}^\delta{}_\alpha \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta + \overset{(4)}{\mathcal{K}}_2 \partial_\chi \mathcal{K}^\delta{}_\beta{}^\delta{}_\alpha \partial^\chi \mathcal{K}^\alpha{}_\alpha{}^\beta{}_\beta + \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^\delta{}_\beta - 2 \overset{(4)}{\mathcal{K}}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}{}^\delta{}_\alpha - \\ \overset{(4)}{\mathcal{K}}_3 \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^\delta{}_\beta - \frac{1}{2} \overset{(4)}{\mathcal{K}}_3 \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}^\delta{}_\beta{}^\delta{}_\chi + \overset{(4)}{\mathcal{K}}_{15} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2 \overset{(4)}{\mathcal{K}}_{15} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_1^{\#1\alpha} + 2 \mathcal{J}_1^{\#2\alpha} == 0$	3	$\partial_\chi \partial^\alpha \mathcal{J}^\beta{}_\beta{}^\chi{}_\chi + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta == 3 \partial_\chi \partial_\beta \mathcal{J}^{\beta\alpha\chi}$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == \mathcal{J}_{1^+}^{\#2\alpha\beta}$	3	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} +$ $4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^\delta{}_\epsilon{}^\epsilon + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}{}_\delta ==$ $3 \partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3 \partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_\delta + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} +$ $2 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4 \partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^\delta{}_\delta{}^\epsilon + 3 \eta^{\beta\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}{}_\delta$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}^\chi{}_\chi{}^\delta{}_\delta == 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}^\chi{}_\chi{}^\delta{}_\delta + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	16		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	1	$-\frac{\overset{(2)}{\mathcal{K}}_3}{\overset{(4)}{\mathcal{K}}_1 + \overset{(4)}{\mathcal{K}}_2}$	$-\frac{2}{3 (\overset{(4)}{\mathcal{K}}_1 + \overset{(4)}{\mathcal{K}}_2)}$
	1	$-\frac{\overset{(2)}{\mathcal{K}}_1}{\overset{(4)}{\mathcal{K}}_{15}}$	$-\frac{1}{3 \overset{(4)}{\mathcal{K}}_{15}}$
	3	$\frac{2 \overset{(2)}{\mathcal{K}}_1}{-2 \overset{(4)}{\mathcal{K}}_{15} + \overset{(4)}{\mathcal{K}}_3}$	$\frac{2}{6 \overset{(4)}{\mathcal{K}}_{15} - 3 \overset{(4)}{\mathcal{K}}_3}$
	3	$-\frac{\overset{(2)}{\mathcal{K}}_3}{\overset{(4)}{\mathcal{K}}_2}$	$\frac{2}{3 \overset{(4)}{\mathcal{K}}_2}$
Resolved unitarity condition(s):		(Demonstrably impossible)	